

Introduction of Machine Learning & it's Types

Presenter: Siva Bharath

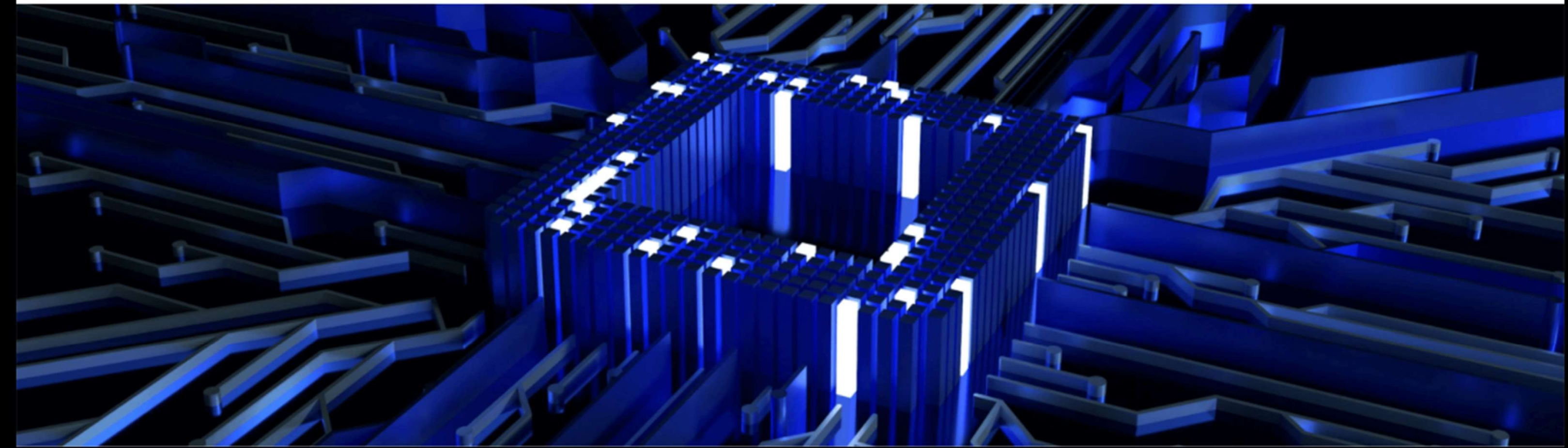
Date: 2026.5.26



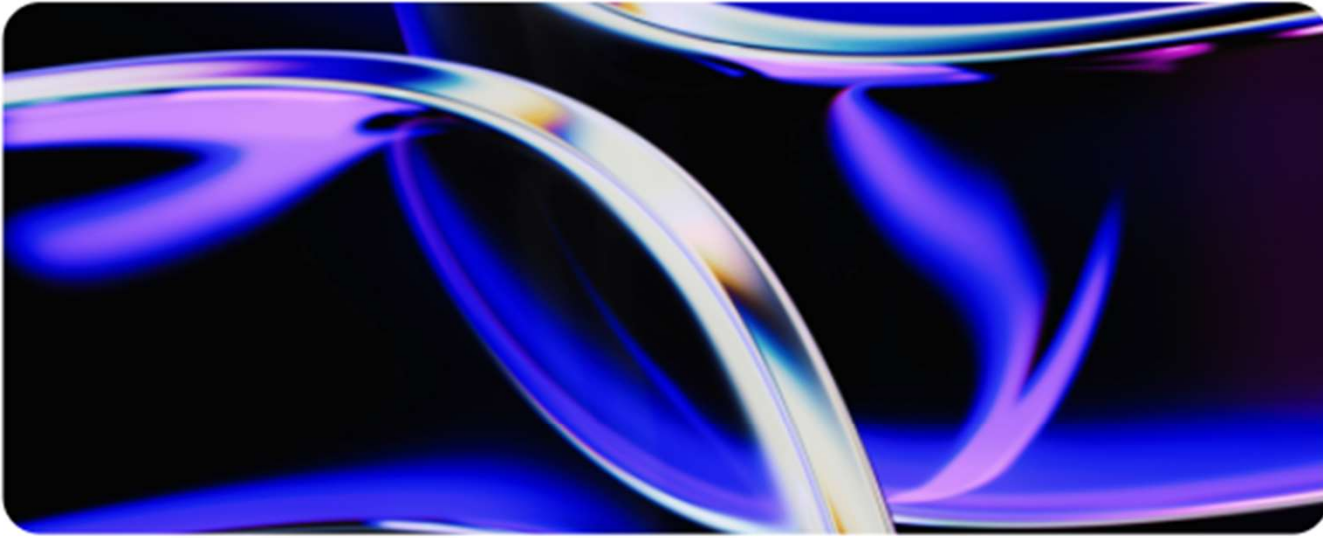
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01 Overview & Objectives



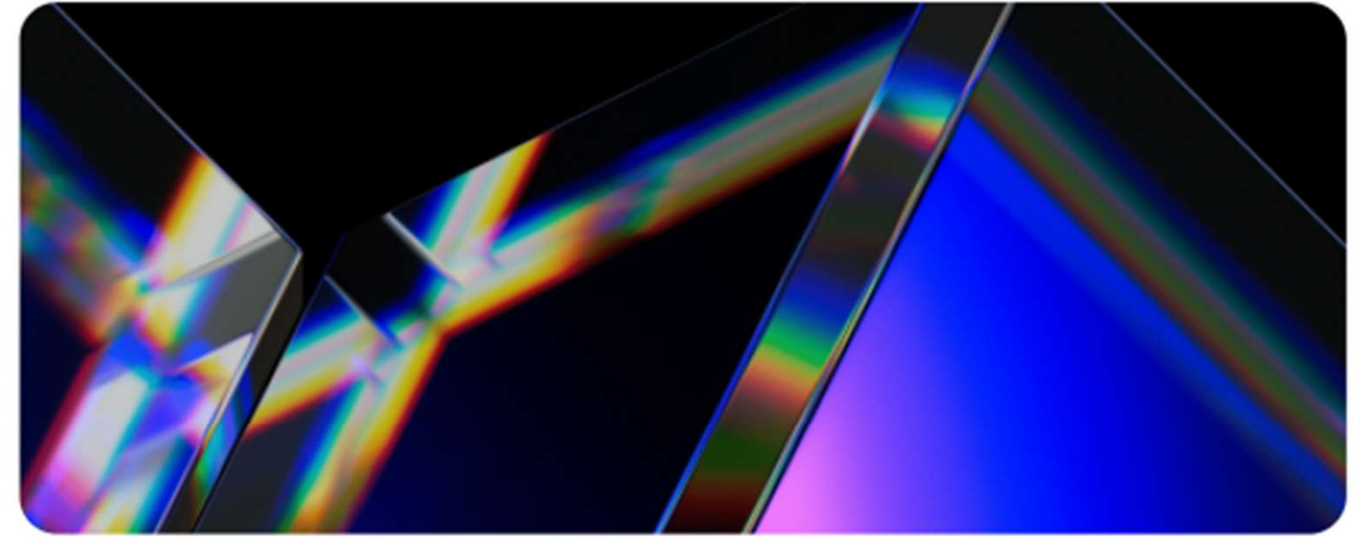
Title & Visual Theme



Typography & Text Layout

Center “Machine Learning” as bold main title, with smaller subtitle “Introduction and Types of Machine Learning” beneath, both using modern sans-serif fonts for clarity.

Align text and visuals symmetrically, keeping hierarchy clear between title, subtitle, and footer, so viewers instantly grasp topic focus and session context.



Cover Design Concept

Use blue-purple gradient with circuit or robotic tech motifs, adding a subtle AI silhouette or hologram illustration to convey modern, futuristic machine learning visuals.

Maintain generous whitespace, few high-impact icons, and a clean footer for presenter name, course, and date, ensuring immediate professionalism and a strong first impression.

ML Goals

Concept Focus

What ML Is

Explain machine learning as systems that learn patterns from data, and clarify how it fits inside the broader field of artificial intelligence today.

ML vs AI

Show how machine learning powers many AI systems, while AI includes broader goals like reasoning, planning, and understanding beyond pure pattern learning.

Why It Matters

Connect machine learning to everyday tools such as recommendations, medical support, and smart assistants, emphasizing its growing impact and relevance.

VS

Method Focus

Types of ML

Introduce supervised, unsupervised, and reinforcement learning using a simple visual comparison to show how each learns from data or feedback.

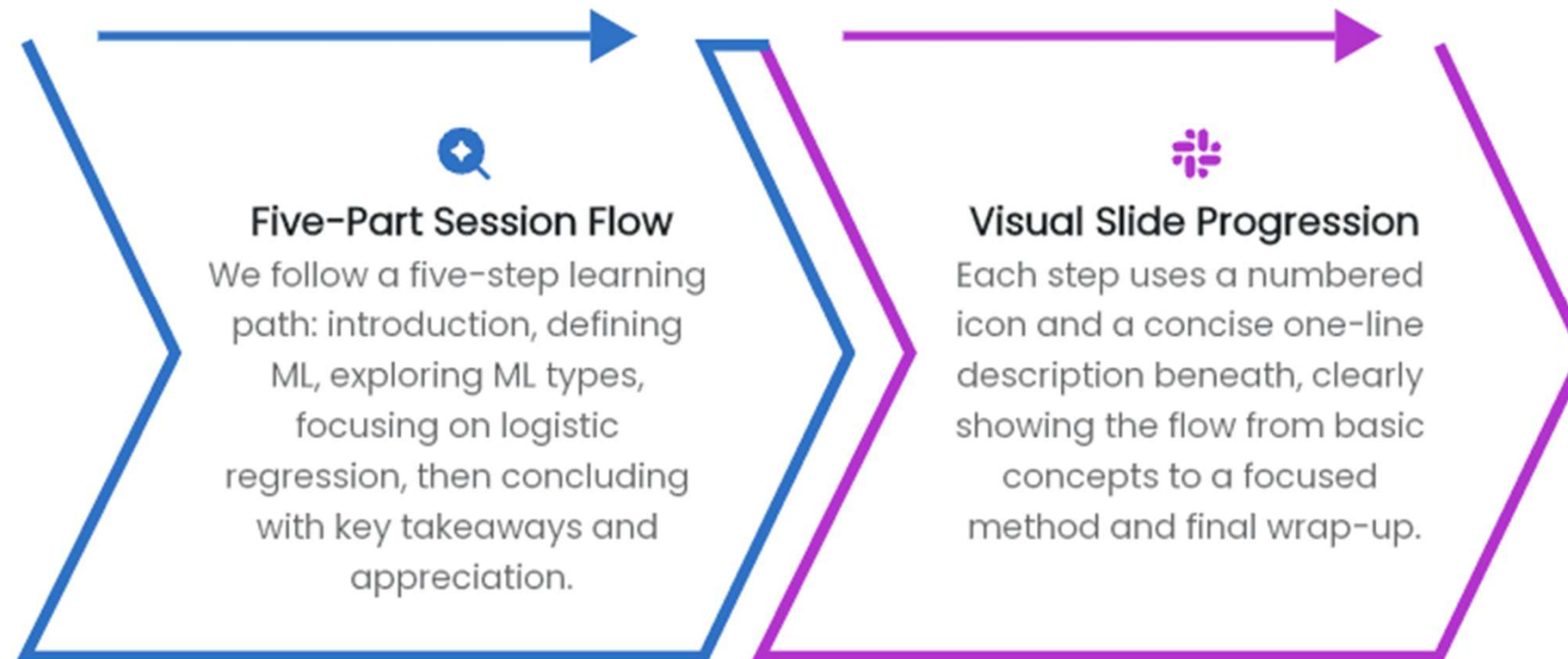
Logistic Regression

Present logistic regression as a core classification algorithm that links inputs to probabilities, building intuition for more advanced machine learning models.

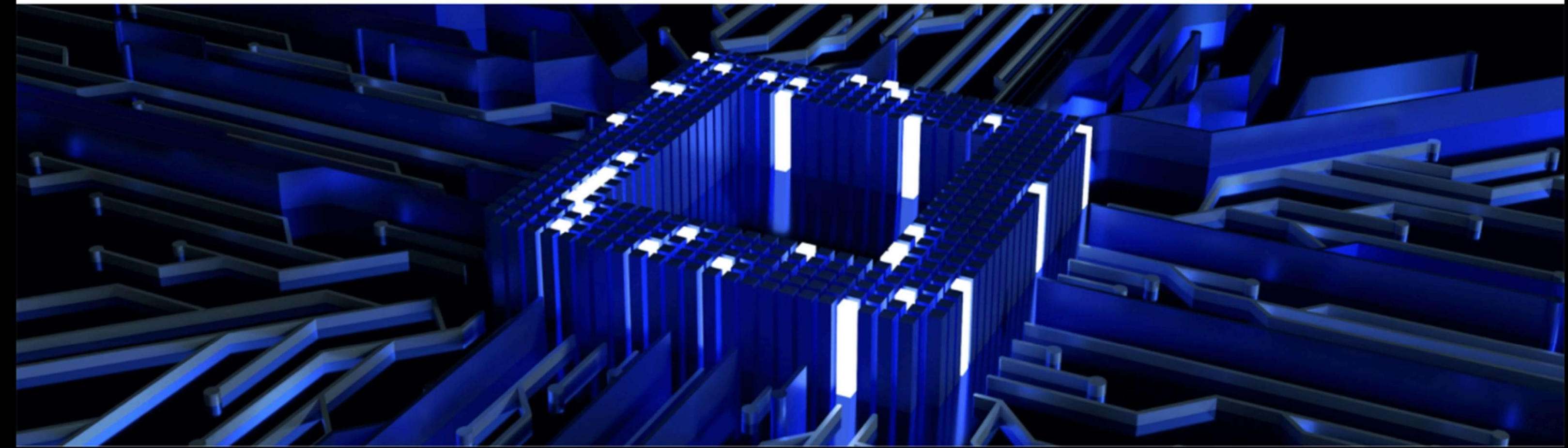
Future Scope

Summarize how machine learning is shaping future work, research, and society, while noting the need for ethics, responsibility, and critical thinking.

Agenda & Structure



02 Foundations of Machine Learning



Machine Learning Essentials



Machine Learning: Introduction

Explore the foundations of machine learning with a bold title slide featuring futuristic AI visuals, highlighting core concepts, objectives, and the evolving role of algorithms in modern technology.



Types and Presentation Details

Introduce supervised, unsupervised, and reinforcement learning, supported by brain-chip icons, data-node graphics, presenter information, and subtle fade-in animations enhancing clarity, professionalism, and audience engagement.

Understanding Machine Learning Basics

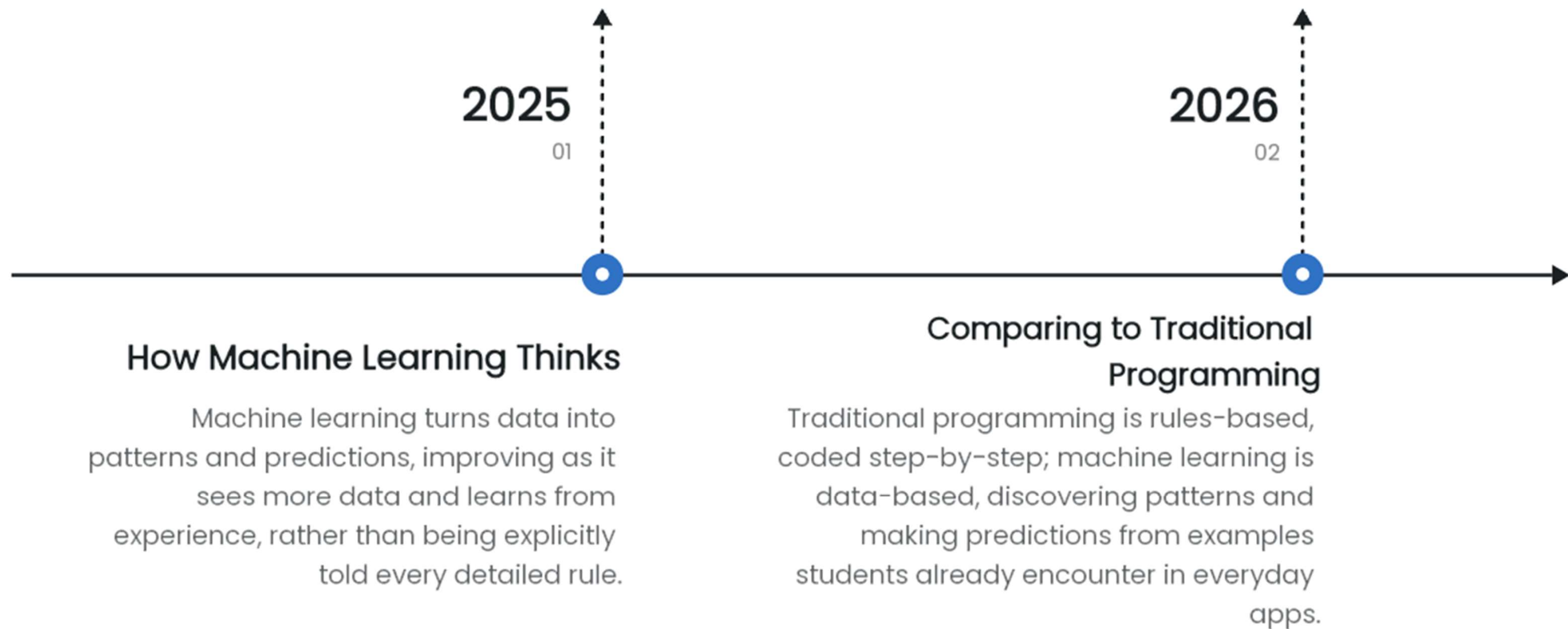
What Is Machine Learning?

Machine Learning is a branch of Artificial Intelligence where computers learn patterns from data to make predictions or decisions, instead of following only explicitly programmed rules.

Key Idea And Practical Examples

Systems learn from data without hand-coding every rule; examples include spam detection, face recognition, and recommendation systems, often illustrated as data flowing into models producing predictions.

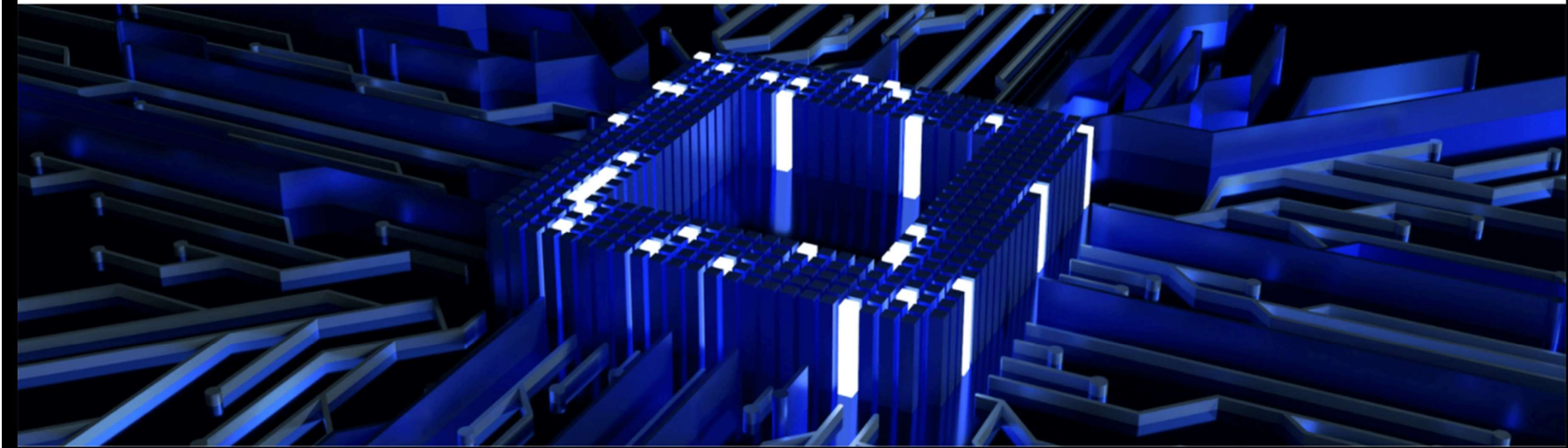
Core Concept Highlights



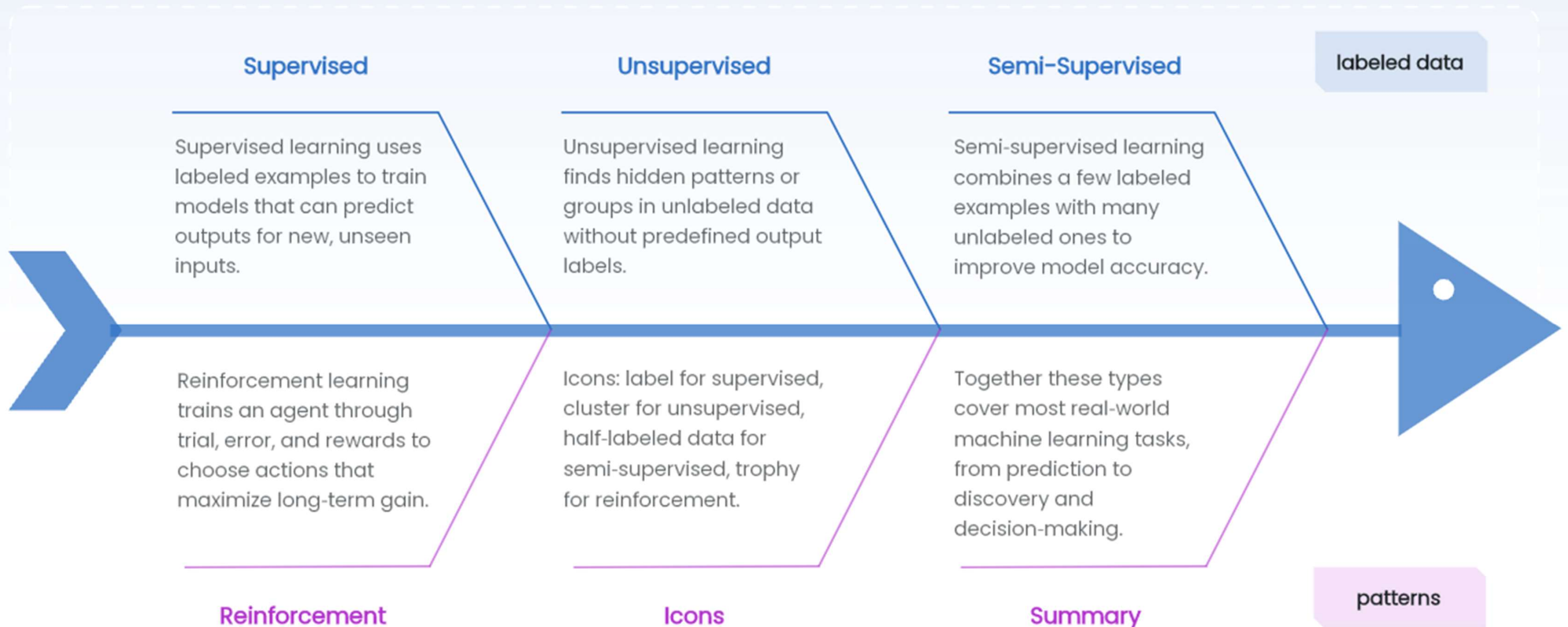


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Types of Machine Learning

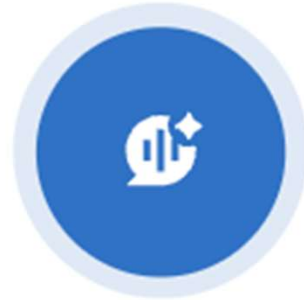


Types of Machine Learning



A quick overview of how machines learn from data using four core learning approaches.

Visual Comparison Diagram



Supervised vs Unsupervised Learning

Supervised learning uses labeled examples, pairing inputs with correct outputs. Unsupervised learning analyzes unlabeled data, discovering hidden structures, clusters, and patterns automatically.



Semi-Supervised vs Reinforcement Learning

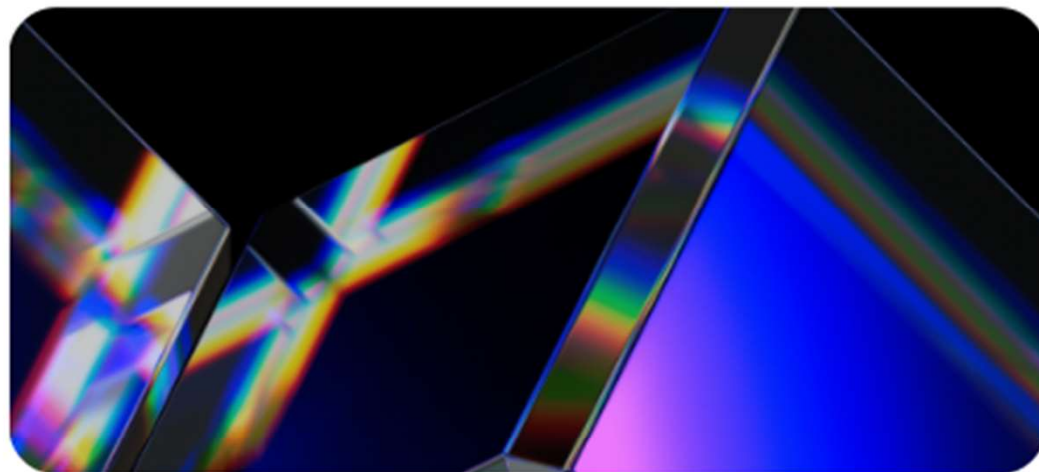
Semi-supervised learning combines few labeled and many unlabeled samples. Reinforcement learning improves through trial-and-error, guided by rewards instead of explicit labels or answers.

Learning Scenarios Examples



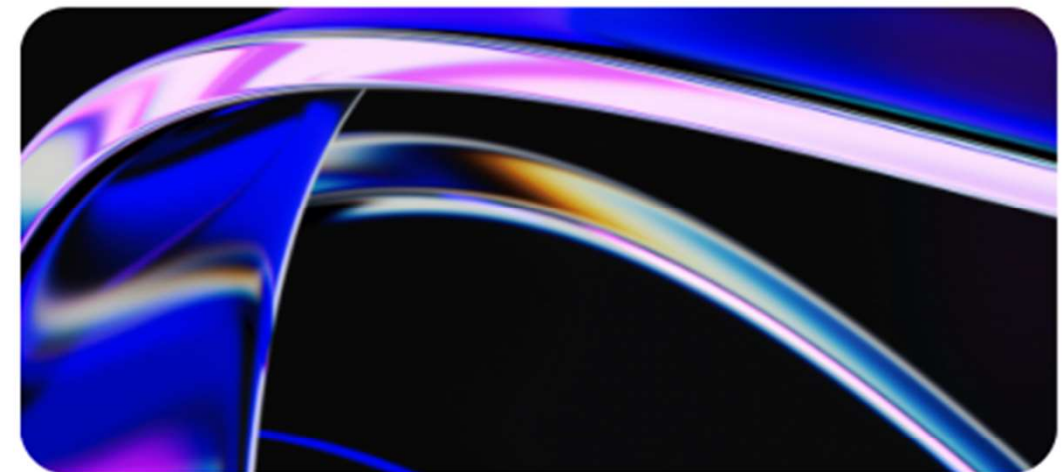
Supervised Learning Scenario

Visual: inbox with labeled emails; caption: "Model learns from spam vs non-spam labels to classify new messages accurately in future, improving email filtering performance over time."

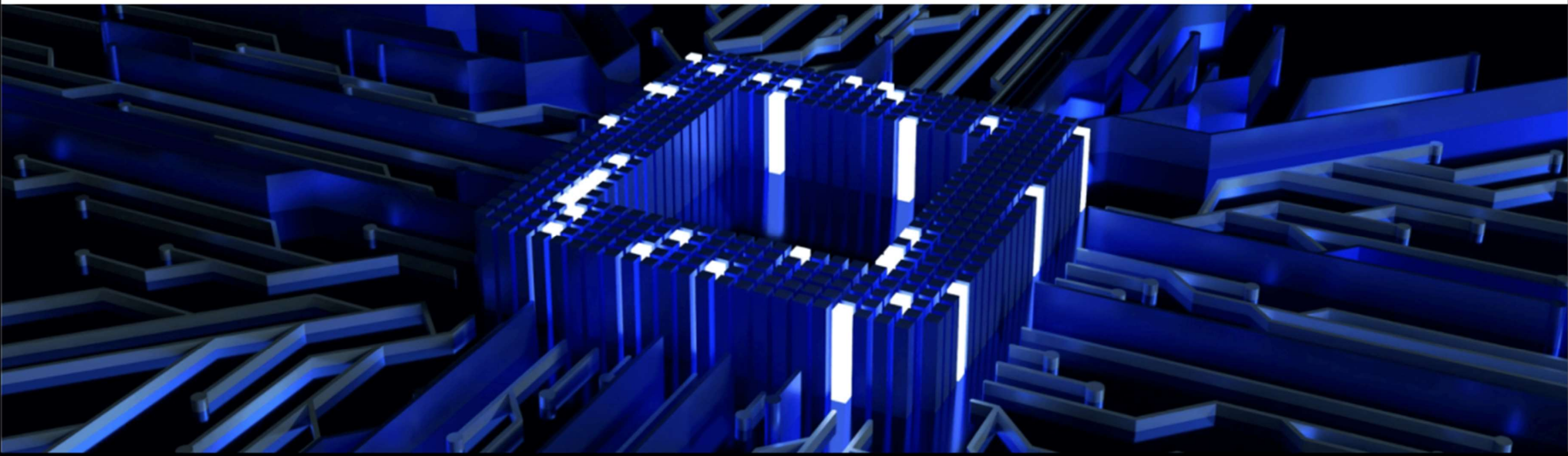


Unsupervised & Reinforcement Scenarios

Visual 1: clustered customer icons; caption: "Algorithm groups customers by behavior without labels, revealing hidden segments." Visual 2: game controller; caption: "Agent improves decisions through rewards and penalties."



04 Logistic Regression Focus



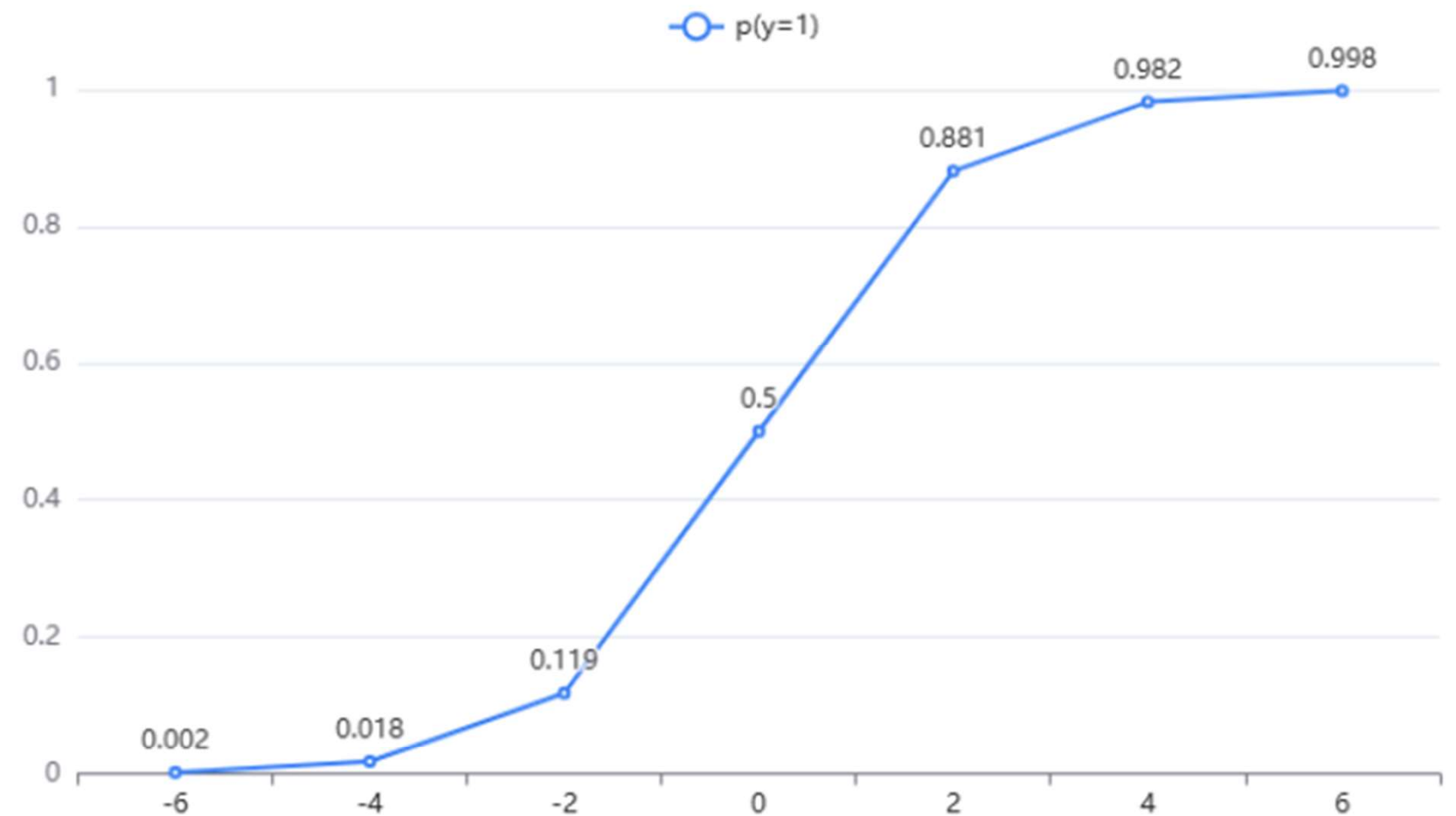
Logistic Regression Basics

Intuition and Use Cases

Logistic regression is a machine learning algorithm mainly used for classification, especially binary decisions like yes or no, spam or not spam, labeled as 0 or 1.

Probability Output and Sigmoid Idea

Instead of predicting a raw number, it outputs the probability of $y=1$ between 0 and 1, using an S-shaped sigmoid curve to map inputs smoothly into probability scores.



Sigmoid Function & Output

Visualizing the Sigmoid Curve

Center image shows an S-shaped sigmoid curve with input scores on the horizontal axis and predicted probability on the vertical axis, emphasizing continuous transformation.

Understanding Probability Outputs

Low input scores yield probabilities near zero, high scores approach one, and a highlighted threshold around zero point five illustrates how the model converts continuous scores into class decisions.



Logistic Regression Applications



Key real-world uses

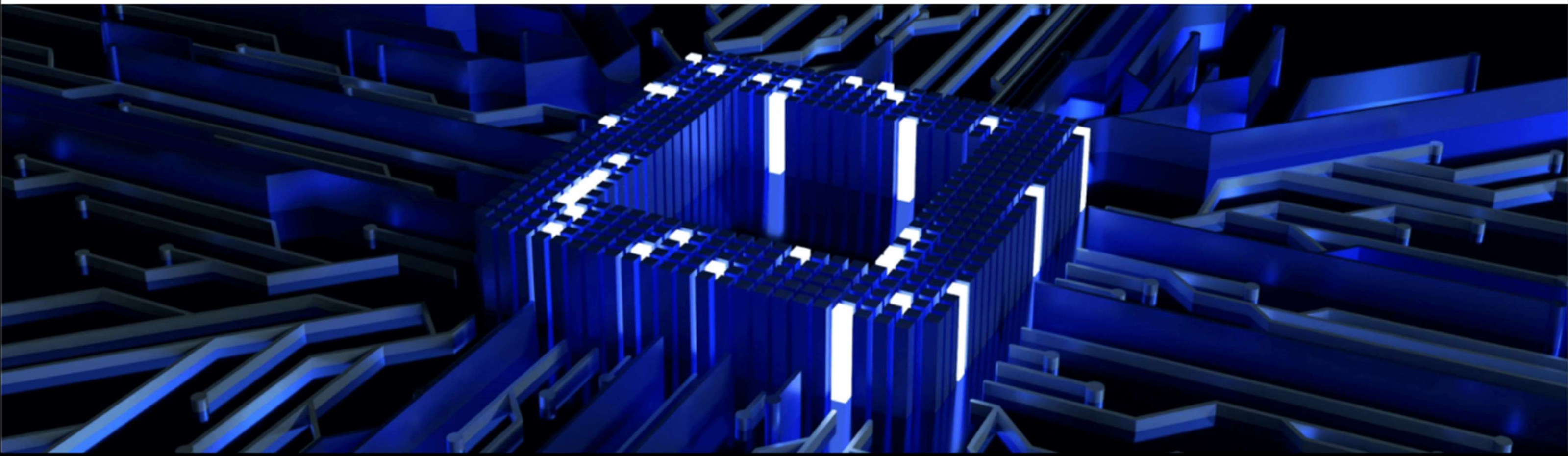
Spam detection for email filtering
Disease prediction from clinical indicators
Customer churn prediction in subscription services



Why it matters

Interpretable, fast, and easy to implement
Strong baseline for many classification tasks
One of the simplest and most widely used ML models

05 Impact & Wrap-Up



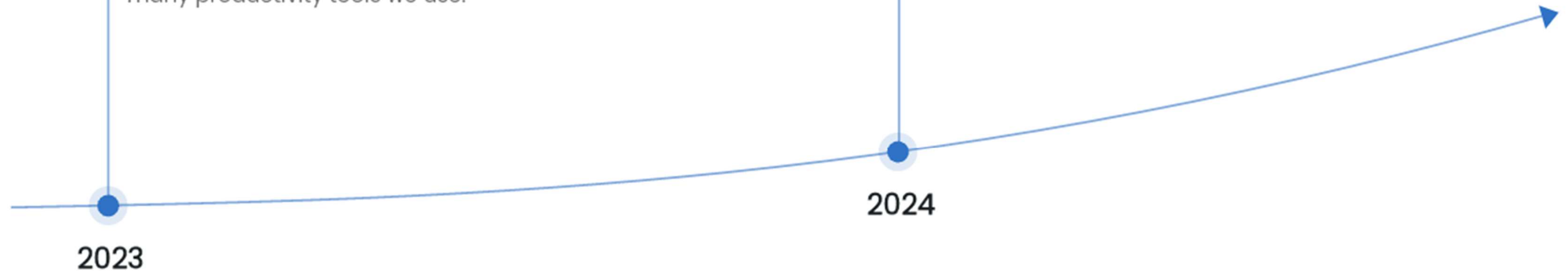
Conclusion & Real-World Importance

● Everyday Impact of Machine Learning

Machine learning powers common technologies, automates routine decisions, and supports smarter, data-driven solutions that appear in search, recommendations, personalization, and many productivity tools we use.

● Future Scope and Responsible Adoption

Real-world applications span business, healthcare, finance, and entertainment, with growing AI integration in daily life and an increasing need for ethical, transparent, and responsibly governed machine learning.



Thank You & Visual Finish



Visual Design & Layout

Large “Thank You” in modern typography centered on slide, AI-themed background with robots, neural networks, or brain-chip imagery, maintaining strong color contrast and generous negative space.

Footer & Animation Details

Add small “Questions?” footer with chat bubble icon, maintain minimal text, suggest subtle outro animation like slow fade-out or gentle slide-up to ensure a polished, professional closing impression.